

Disconnected qTMD/PDF One-Point Building Blocks

Stochastic Quark Loops for Pion and Proton qTMD Analysis

PyQUDA Measurement Notes

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1 Purpose

This note documents the first PyQUDA implementation of disconnected qTMD/PDF one-point quark loops. The source file is `pyquda_measurement_utils/Disconnected_1pt_qTMD_vibe_develop.py`, and the Perlmutter application is `application/qTMD_disconnected_1pt/perlmutter`.

The loops are hadron independent. Pion and proton information enters later, when the loop is combined with the corresponding hadron two-point function in ensemble analysis.

2 Disconnected qTMD Loop

For a stochastic source ξ_r , define

$$\eta_r = D^{-1}\xi_r, \tag{1}$$

where D is the same clover Dirac operator used by the connected qTMD measurements. The implemented disconnected qTMD/PDF loop is

$$L_{\Gamma,b}(q, \tau) = \frac{1}{N_r} \sum_{r=1}^{N_r} \sum_{\mathbf{x}} e^{i\mathbf{q}\cdot\mathbf{x}} \text{Tr}_{sc} \left[\eta_r^\dagger(x) \Gamma O_b \xi_r(x) \right], \quad (2)$$

where $x = (\mathbf{x}, \tau)$, Γ is one of the 16 bilinear gamma matrices, and O_b is a nonlocal displacement or Wilson-line operator.

The current code convention is therefore

$$\boxed{\text{loop convention} = \eta^\dagger \Gamma O_b \xi}. \quad (3)$$

The nonlocal operator O_b acts on the stochastic source side. This choice is intentionally minimal and makes the first implementation easy to compare against connected PDF and local-current limits.

3 Supported Operators

The implementation supports four operator kinds:

operator kind	Operator	Status
GI_PDF	Straight- z gauge-invariant PDF Wilson line	Implemented
CG_PDF	Straight- z coordinate-gauge displacement	Implemented
CG_qTMD	qTMD-style coordinate-gauge displacement	Implemented
GI_qTMD	Fixed-length gauge-invariant staple	Implemented

3.1 Straight- z PDF Operators

For PDF operators,

$$b = b_z \hat{z}. \quad (4)$$

The coordinate-gauge version is a plain lattice shift:

$$O_b^{\text{CG-PDF}} \xi(x) = \xi(x + b_z \hat{z}). \quad (5)$$

The gauge-invariant version applies the straight Wilson line through repeated covariant shifts:

$$O_b^{\text{GI-PDF}} \xi(x) = U_z(x) U_z(x + \hat{z}) \cdots \xi(x + b_z \hat{z}), \quad (6)$$

with the corresponding backward-link product for negative b_z . In code this is implemented by repeated calls to `gauge.pure_gauge.covDev`.

3.2 Coordinate-Gauge qTMD Operator

The current qTMD operator follows the connected qTMD diagnostic convention:

$$O_b^{\text{CG-qTMD}} \xi(x) = \xi(x + b_T \hat{e}_\perp + b_z \hat{z}), \quad (7)$$

where $\hat{e}_\perp$ is either $\hat{x}$ or $\hat{y}$. The two transverse directions are stored as `b_X` and `b_Y` in the HDF5 output.

3.3 Gauge-Invariant qTMD Staple Operator

The production code also supports a fixed-length gauge-invariant staple operator, `operator kind=GI_qTMD`. Its Wilson index is $[b_T, b_z, \eta, \perp]$, with even physical b_z , non-negative η , and $\eta \geq |b_z|/2$.

4 Momentum Convention

The loop momentum is the current momentum transfer,

$$q = p_f - p_i. \quad (8)$$

The one-point loop itself has no hadron source or sink momentum. Those labels enter when the loop is combined with a pion or proton two-point function.

The code uses `pyquda_utils.phase.MomentumPhase`, whose convention is

$$\Phi_q(x; x_0) = \exp \left[+2\pi i \sum_{j=x,y,z} \frac{q_j(x_j - x_{0,j})}{L_j} \right]. \quad (9)$$

The disconnected loop is projected as

$$L_{\Gamma,b}(q, \tau) = \sum_{\mathbf{x}} \Phi_q(\mathbf{x}; \mathbf{x}_0) \ell_{\Gamma,b}(\mathbf{x}, \tau), \quad (10)$$

so a local test field with plane-wave dependence

$$\ell_{\Gamma,b}(\mathbf{x}, \tau) \propto \exp \left[-2\pi i \sum_{j=x,y,z} \frac{q_j(x_j - x_{0,j})}{L_j} \right] \quad (11)$$

is selected by the $+q$ projection. This is the same Fourier sign used in the connected qTMD/EMT application layers; the physical transfer label remains $q = p_f - p_i$.

5 Combination with Hadron Two-Point Functions

For a hadron two-point correlator $C_2^H(p_f, t)$, the disconnected qTMD three-point building block is formed as

$$C_{3,\Gamma,b}^{H,\text{disc}}(p_f, p_i; t, \tau) = \langle C_2^H(p_f, t) L_{\Gamma,b}(q, \tau) \rangle_{\text{cfg}} - \langle C_2^H(p_f, t) \rangle_{\text{cfg}} \langle L_{\Gamma,b}(q, \tau) \rangle_{\text{cfg}}, \quad (12)$$

with

$$q = p_f - p_i. \quad (13)$$

The corresponding ratio is typically

$$R_{\Gamma,b}^{H,\text{disc}}(t, \tau) = \frac{C_{3,\Gamma,b}^{H,\text{disc}}(p_f, p_i; t, \tau)}{C_2^H(p_f, t)}, \quad (14)$$

before applying the qTMD/PDF renormalization and soft-factor analysis.

The vacuum-subtraction term is essential because the one-point loop is hadron independent.

6 Stochastic Sources and Hierarchical Probing

The source scheme is controlled by

$$\text{noise_scheme} \in \{\text{zn}, \text{hierarchical_probing}\}. \quad (15)$$

For hierarchical probing, a base source ξ_b is multiplied by Rademacher probing vectors $h_k(x) = \pm 1$:

$$\xi_{b,k}(x) = h_k(x) \xi_b(x). \quad (16)$$

The number of effective inversions is

$$N_{\text{eff}} = N_{\text{base}} N_{\text{HP}}. \quad (17)$$

The implemented HP ordering choices are

Ordering	Meaning
<code>global_xyz_t_gray_projected_to_evenodd</code>	Gray code the full x, y, z, t site id.
<code>spatial_xyz_then_t_gray_projected_to_evenodd</code>	Gray code spatial sites first, then append time bits.

Spin-color dilution and time dilution are not currently implemented.

The default base field is decomposition-independent full-volume $\mathbb{Z}_4$ noise. Its fixed counter contains the global four-dimensional coordinate, spin, color, gauge-configuration number, base-noise index, and an independent stream salt. Therefore any base or HP interval can be regenerated without consuming the earlier random-number sequence. Using the same backend RNG seed independently on equal-shaped MPI sublattices would instead repeat the same rank-local field, violate the required off-diagonal noise covariance, and make the estimator decomposition dependent. Adding the rank to the seed is not sufficient for cross-decomposition reproducibility; the global-coordinate counter is the production solution.

6.1 Base-Oriented Production and Finalization

Production uses base/HP-interval shards by default. One complete base estimator is

$$L_b = \frac{1}{N_{\text{HP}}} \sum_{h=0}^{N_{\text{HP}}-1} L_{b,h}. \quad (18)$$

Part files contain at most 64 solves by default. A partial HP part is a checkpoint only; a base completion marker is atomically published after every part has been reopened and its metadata, shape, and source indices validated. Compatible parts are skipped on resume, while incompatible or corrupt parts cause an error.

Measurement jobs do not publish the canonical loop. A separate serial finalizer requires complete coverage of all configured bases and streams each part into a temporary canonical HDF5 file while accumulating the average without retaining the full source axis. Only after successful validation and writeout is the canonical file atomically replaced. Non-overlapping base ranges can be scheduled independently; overlapping jobs are unsupported.

7 HDF5 Layout

New output uses the source-independent helper `get_disconnected_qTMD_loop_file_tag` and canonical name

$$\text{qTMD1pt}/\langle\text{lat}\rangle.\text{qTMD1pt}.\langle\text{cfg}\rangle.\langle\text{ama}\rangle.\langle\text{sm}\rangle.\text{h5}.$$

The old source-position helper remains available for existing files and APIs; the full-volume loop itself has no hadron source position. The writer stores:

$$\text{raw/loop_pervec} \quad \text{with shape} \quad [N_{\text{eff}}, N_W, N_\Gamma, N_q, N_t]. \quad (19)$$

The raw bookkeeping datasets are

$$\text{raw/source_index}, \quad \text{raw/base_noise_index}, \quad \text{raw/hp_index}. \quad (20)$$

The averaged data are divided by the spatial volume and stored in a qTMD-like layout:

$$\text{avg/SS}/\langle\text{gamma}\rangle/\text{PX}...\text{PY}...\text{PZ}.../\text{b_X}/\text{eta0}/\text{bT}.../\text{bz}.... \quad (21)$$

For y -direction transverse displacements the group is `b.Y`. All serial HDF5 files are opened only by MPI rank zero. Because no fermion flow changes the resident gauge in this workflow, the Dirac gauge is loaded once before the source loop and reused.

8 Validation Status

The first implementation has been checked with:

- Python syntax compilation.
- A Perlmutter login-node smoke test for `GI_PDF` at $b_z = 0$.
- A Perlmutter login-node smoke test for `GI_PDF` at $b_z = 1$.
- A Perlmutter login-node smoke test for `CG_qTMD` at $b_T = 1, b_z = 1$.

8.1 Local/PDF Limit Sanity Checks

At zero displacement and zero momentum transfer, the different disconnected operator implementations should collapse to the same local bilinear loop:

$$O_{b_z=0}^{\text{GI_PDF}} = O_{b_z=0}^{\text{CG_PDF}} = O_{b_T=0, b_z=0}^{\text{CG_qTMD}} = 1. \quad (22)$$

Therefore, for the same gauge field, stochastic source, gamma matrix, and momentum $q = 0$, the expected equality is

$$L_{\Gamma, b_z=0}^{\text{GI_PDF}}(0, \tau) = L_{\Gamma, b_z=0}^{\text{CG_PDF}}(0, \tau) = L_{\Gamma, b_T=0, b_z=0, \hat{x}}^{\text{CG_qTMD}}(0, \tau) = L_{\Gamma, b_T=0, b_z=0, \hat{y}}^{\text{CG_qTMD}}(0, \tau). \quad (23)$$

This check is useful because it tests the HDF5 bookkeeping, Wilson-index ordering, gamma loop ordering, and the coordinate-gauge/GI local limits without requiring a hadron two-point function.

The current S8T32 Perlmutter login-node check used one matched counter-based stochastic vector and found exact agreement:

$$\max |L^{\text{GI_PDF}} - L^{\text{CG_PDF}}| = \max |L^{\text{GI_PDF}} - L_{\hat{x}}^{\text{CG_qTMD}}| = \max |L^{\text{GI_PDF}} - L_{\hat{y}}^{\text{CG_qTMD}}| = 0. \quad (24)$$

The same equality was verified for both `raw/loop_pervec` and the averaged $\Gamma = \gamma_5$ HDF5 path `avg/SS/5/PX0PY0PZ0/b_X/eta0/bT0/bz0`, with the y -direction branch stored under `b_Y` for `CG_qTMD`.

The regression helper is `tests/test_qtmd_disconnected_local_pdf_limit.py`. It skips automatically if the local HDF5 smoke-test files are not present, and otherwise requires exact equality for the local/PDF limit above.

8.2 Nonzero- b_z Shift Sanity Check

A second check probes the straight- z displacement convention at nonzero separation. For the coordinate-gauge operators, the PDF path at $b_T = 0$ should be exactly the $b_T = 0$ limit of the qTMD coordinate shift:

$$O_{b_z=\pm 1}^{\text{CG_PDF}} = O_{b_T=0, b_z=\pm 1, \hat{x}}^{\text{CG_qTMD}} = O_{b_T=0, b_z=\pm 1, \hat{y}}^{\text{CG_qTMD}}. \quad (25)$$

This equality checks the sign of the z -shift, the Wilson-index ordering, and the transverse-branch bookkeeping. The corresponding S8T32 check found exact agreement for both $b_z = +1$ and $b_z = -1$:

$$\max |L_{\Gamma, b_z=\pm 1}^{\text{CG_PDF}} - L_{\Gamma, b_T=0, b_z=\pm 1, \hat{x}/\hat{y}}^{\text{CG_qTMD}}| = 0. \quad (26)$$

The gauge-invariant PDF loop uses the same Wilson-index labels, $[0, 0, 0, 0]$, $[0, +1, 0, 0]$, and $[0, -1, 0, 0]$, but it is not expected to equal the coordinate-gauge loop on a nontrivial gauge field because the straight Wilson line is active:

$$O_{b_z=\pm 1}^{\text{GI_PDF}} = U_z \cdots \xi(x \pm \hat{z}) \neq \xi(x \pm \hat{z}) = O_{b_z=\pm 1}^{\text{CG_PDF}}. \quad (27)$$

The nonzero difference between `GI_PDF` and `CG_PDF` therefore confirms that the GI path is not accidentally falling back to the plain coordinate shift.

The regression helper is `tests/test_qtmd_disconnected_nonzero_bz.py`. During this check it also exposed and fixed a transverse-branch bug in the `CG_qTMD` operator: when switching from the $\hat{x}$ branch to the $\hat{y}$ branch, the shifted stochastic source must be reset to the local source before applying the new branch displacement.

8.3 Nonzero- q Phase Sanity Check

The regression helper `tests/test_disconnected_momentum_phase.py` uses a pure NumPy toy field to check the nonzero-momentum Fourier sign. For a tiny lattice and $q = (1, -2, 3)$, it verifies

$$\sum_{\mathbf{x}} \Phi_q(\mathbf{x}; 0) \Phi_q^*(\mathbf{x}; 0) = L_x L_y L_z, \quad (28)$$

while the opposite projection vanishes:

$$\sum_{\mathbf{x}} \Phi_{-q}(\mathbf{x}; 0) \Phi_q^*(\mathbf{x}; 0) = 0. \quad (29)$$

The same test also checks that changing x_0 only changes the expected overall phase, not the selected momentum magnitude.

9 Gauge-Invariant Staple qTMD Design

The `CG_qTMD` operator is only a coordinate-gauge diagnostic. The production `GI_qTMD` operator replaces the plain shift by a staple Wilson line. The convention is the fixed-staple-length convention used by the legacy GPT proton qTMD workflow. In the current notation, b_z is the physical final longitudinal separation and must be an even integer:

$$O_{b, \eta, \perp}^{\text{GI-qTMD}} \xi(x) = U_{\mathcal{C}(x; b, \eta, \perp)} \xi(x + b_T \hat{e}_\perp + b_z \hat{z}), \quad (30)$$

with

$$b_z \in 2\mathbb{Z}, \quad \eta \geq \frac{|b_z|}{2}. \quad (31)$$

The three-segment staple path is

$$\mathcal{C}(x; b, \eta, \perp) : x \rightarrow x + \left(\eta + \frac{b_z}{2}\right) \hat{z} \rightarrow x + \left(\eta + \frac{b_z}{2}\right) \hat{z} + b_T \hat{e}_\perp \rightarrow x + b_z \hat{z} + b_T \hat{e}_\perp. \quad (32)$$

The signed segment lengths are

$$\left[\left(z, \eta + \frac{b_z}{2}\right), (\perp, b_T), \left(z, \frac{b_z}{2} - \eta\right) \right]. \quad (33)$$

For $\eta \geq |b_z|/2$, the geometric Wilson-line length is independent of b_z :

$$\left| \eta + \frac{b_z}{2} \right| + b_T + \left| \eta - \frac{b_z}{2} \right| = 2\eta + b_T. \quad (34)$$

This fixed length is useful for renormalization because Wilson-line self-energy effects are matched across different b_z at fixed η and b_T . With $b_T = 0$ and $\eta = |b_z|/2$, the staple reduces to the straight `GI_PDF` Wilson line. At $b_T = b_z = \eta = 0$, it reduces to the local operator.

The legacy GPT implementation used an equivalent convention with b_z^{legacy} denoting half of the final longitudinal separation:

$$b_z^{\text{current}} = 2b_z^{\text{legacy}}. \quad (35)$$

Its path comments and implementation correspond to $(\eta + b_z^{\text{legacy}}) + b_T - (\eta - b_z^{\text{legacy}})$, which is the same fixed-length staple after this relabeling.

The `GI_qTMD` implementation keeps the existing `CG_qTMD`, `CG_PDF`, and `GI_PDF` outputs unchanged. The first code-side helper is `gi_qtmd_staple_segments(W_index)`, which maps $[b_T, b_z, \eta, \perp]$ to the signed path segments

$$\left[\left(z, \eta + \frac{b_z}{2} \right), (\perp, b_T), \left(z, \frac{b_z}{2} - \eta \right) \right]. \quad (36)$$

The helper is covered by `tests/test_disconnected_gi_qtmd_staple_design.py`; it fixes the local, straight-PDF, fixed-length generic path, and invalid-index conventions.

The reference code-side helper is `create_fermion_TMD_GI(gauge, fermion, W_index)`. It applies the full staple path from the original local fermion field for each W -index, using repeated `gauge.pure_gauge.covDev` calls.

For production, the default `QTMD_1PT_GI_STAPLE_MODE=link_cache` first builds gauge-only staple transporters with `build_gi_qtmd_staple_links`. The construction converts an identity color matrix field into color-column fermions, applies the same `covDev` path, and converts the result back to a link field. This keeps the transporter convention identical to the reference direct-covariant derivative path while allowing all stochastic sources on the same gauge configuration to reuse the same staple links. The debug mode `QTMD_1PT_GI_STAPLE_MODE=direct_covdev` keeps the older source-by-source reference implementation.

The link-cache implementation is tested by:

1. identity-gauge equality with coordinate shifts;
2. nontrivial S8T32 test-gauge equality between cached links and direct `covDev`;
3. the PDF limit $\text{GI_qTMD}(b_T = 0, \eta = |b_z|/2) = \text{GI_PDF}(b_z)$;
4. a nonzero-transverse-staple HDF5 comparison at $(b_T, b_z, \eta) = (1, 2, 1)$, where `link_cache` and `direct_covdev` agree at the floating-point roundoff level.

The next physics validation should compare the local/PDF limits against the connected pion and proton qTMD workflows after constructing the disconnected ratio with matched two-point functions.